PataFEST: PotATo crop effective mAnagement strategies to tackle Future pEST threats

List of participants

Nº	Participant Organisation Name	Key Role	Type	Country
1	Fundación para el Desarrollo y la Innovación Tecnológica [**FUND**]	Coordinator. Postharvest biocontrol solutions	RTO	ES
2	Instituto Vasco de Investigación y Desarrollo Agrario [**NEIK**]	Resistance genes identification. Molecular Breeding	RTO	ES
3	Instituto Nacional de Investigaciones Agropecuarias [**INIAP**]	Characterization of germplasm. Field and bioassays	RTO	EC
4	IRIS Technology Solutions [**IRIS**]	Modelling and data processing. Image analysis digital tool developer	SME	ES
5	Kneia SL **[KNEIA]**	Diss, Comm., Exploit & Citizen Science	SME	ES
6	Cranfield University [**CRUN**]	Volatile biomarkers identification. Storage technology. Plant soil management	UNI	UK
7	Udapa S. Coop. [**UDAPA**]	Validator of postharvest strategies	SME	ES
8	Chemia Spa [**CHEM**]	Plant and postharvest solutions manufacturer	SME	IT
9	Palacký University Olomuc [**UPOL**]	Phenotyping study for portable sensor development	UNI	CZ
10	Technion-Israel Institute of Technology [**TECHN**]	Vector repellent and antifungal solution	RTO	IL
11	European Potato Trade Association [**EUPAT**]	Multi-actor approach. Diss&Comm.	ASS	BE
12	As. Agraria de Jóvenes Agricultores [**ASAJA**]	Multi-actor approach. Diss&Comm.	ASS	ES
13	Alma Mater Studiorum Universitá di Bologna [**UNIBO**]	Biobased solutions for soil management	RTO	IT
14	Kartoffel-Boehmer [**BOEH**]	Validator of postharvest strategies	SME	DE
15	European Citizen Science Association [**ECSA**]	Citizen Science approach	ASS	DE

Acronyms: BC: *Bactericera cockerelli;* **CA**: controlled atmosphere*;* **CLso**: *Candidatus Liberibacter solanacearum;* **IPM**: Integrated pest management; **IR**: Infrared; **LC-MS**: liquid chromatography-mass spectrometry; **MB**: Molecular breeding; **PHTS**: Phenotype-based High-throughput screening; **RG**: resistant genes; **SNPs**: single nucleotide polymorphisms; **VOCs**: volatile organic compounds; **ZC**: Zebra chips.

1 EXCELLENCE

1.1 Objectives and ambitions

1.1.1 Objectives

The **potato (*Solanum tuberosum*)** is a key crop in Europe. In 2020, the EU's potato production was 55.3 million tonnes (worth 12.3 billion euros), with Germany (21.2%), Poland (16.4%), France (15.7%), and the Netherlands (12.7%) being the top EU producers. Potato is **vulnerable to a variety of pests and soil-borne pathogen** diseases, resulting in significant economic losses in the potato industry and food losses (in line with SDG 12.3)[1,2]. For example, in 2011, the United States recorded **losses in potato yield of up to 75%,** and in other cases, the damages are so severe that the entire crop is lost[3]. The European regulatory framework for potato's imports and production is very strict. For example, the European and Mediterranean Protection Organization (EPPO) standard for potato establishes recommendations for the international trade to prevent the spread of certain pests [4]. Although there are strict controls, globalisation, climate change and economical drivers are threatening the **entrance of emerging or new potato pests from imported potato to EU**.

Chemical pesticides, on the other hand, are commonly utilised in both preharvest and postharvest methods to manage pest threats and secure yields in plant production. For all that, the **EU Green Deal aims to reduce pesticide use by 50%** through implementing more sustainable plant health solutions[5]. Possible consequences derived from the excessive use of chemical pesticides and the transition towards their substitution are 1) increases **pesticide resistance** in bacteria, insects, and soil-borne pathogens, and 2) **reappearance of soil pathogens**, potentially increasing the incidence of soil infections during postharvest storage.

[1] The EU potato sector – statistics on production, prices and trade. July 2021. Available here.
[2] SDG 12.3: By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses.
[3] Greenway G. (2014) Am. J. Potato Res. (2014) 91:714–719. DOI:10.1007/s12230-014-9404-x.
[4] PM 8/1 (2) Potato. (2017). Bulletin OEPP/EPPO, 47(3), 487–503.
[5] European Green Deal. Available here.

Zebra chip disease

One of the most important **pests in the potato plant** is caused by **bacteria *Candidatus Liberibacter solanacearum* (CLso)** and its disease is known as **zebra chip (ZC)**. The introduction the CLso bacterium on potato is mainly linked to the presence of the **vector psyllid *Bactericera cockerelli* (BC)** causing severe damage both in terms of yield losses and quality. Plant symptoms are related to leaf chlorosis, discoloration, curling and eventually premature plant death. Potato chip industry is also affected by this disease since the CLso-infected potato tubers are often deformed, of low quality, with dark stripes, making chips bitter to taste, and unacceptable for manufacturers to commercialise. CLso has been found throughout North and South America, Canada, New Zealand, and Australia and the disease is reported in other solanaceous crops such as tomato, eggplant, and pepper [6,7]. Although there is currently little risk of the ZC vector entering the European potato seed supply chain, **preventative measures are implemented to decrease the potential appearance of this disease in EU**.

Vector psyllid (BC)

For example, together with traditional application of chemical pesticide, EPPO classified ZC as an **A1 quarantine pest (**pest that is not found in the EPPO territory, but it is recommended to carry out a quarantine management strategy) [8]. However, considering the potential threat of this emerging pest, the development of further management strategies for prevention, early detection and surveillance is required.

In this context, the novel method of **Molecular Breeding (MB)** confers a new dimension to the traditional chemical control disease plan. MB is considered an environmentally safe approach that is focused on the study of the plant genome to detect potential **resistance genes (RG)** capable of recognising invading pathogens and initiating active plant defence mechanisms. Furthermore, MB integrates the identification of diverse molecular DNA plant markers such as simple sequence repeats (SSR), and more commonly single nucleotide polymorphisms (SNPs) that are tightly linked to phenotypic disease. Another relevant analysis in the MB represents quantitative trait loci (QTL) as a statistical method that links both types of information, the phenotypic data and genotypic data derived from the molecular markers to explain the genetic basis of variation. Besides, the identification and characterization of the RG and the detection of functional genetic markers are considered a crucial way to recognize and characterize the disease-resistance mechanisms to improve the **selection of resistant varieties of potato at early stages of breeding**.

Dry-rot infecting potato

As mentioned above, emerging threats will also affect potato safety and quality during the postharvest activities, generating mainly blemish on the potato and make it unmarketable. Main **potato postharvest diseases** include dry rot caused by ***Fusarium sambucinum***, black dot caused by ***Colletothricum coccodes***, and silver scurf provoked by ***Helminthosporium solani***. These pathogens are soil-borne and have incidence in both field and storage stages with dramatic economic losses for the potato farmers and pre-pack potato industry since the postharvest development of such diseases will result in unmarketable product[9]. Standard control for these diseases in warehouses includes elimination of infected tubers prior to storage and storage management using forced air ventilation, and controlled temperature and humidity feedback systems[10]. The use of chlorine-based disinfectants to treat postharvest potato tubers are also applied. However, chlorine may not completely prevent storage pathogens, and the presence of chlorinated residues (chlorates and perchlorates) may cause consumer worries, prompting them to demand more environmentally friendly and chemical-free alternatives, particularly for organic farming.

In view of the current scenario to preserve the health of potato plant and the quality of the final product against its emerging pests, the project PataFEST and its **15 complementary partners** are conceived, including the **international cooperation with Ecuador**, a country affected by the same pests tackled in this project. **PataFEST aim**: **1)** to **research at molecular level the main resistance potato varieties** against CLso and postharvest pathogens, **2)** to

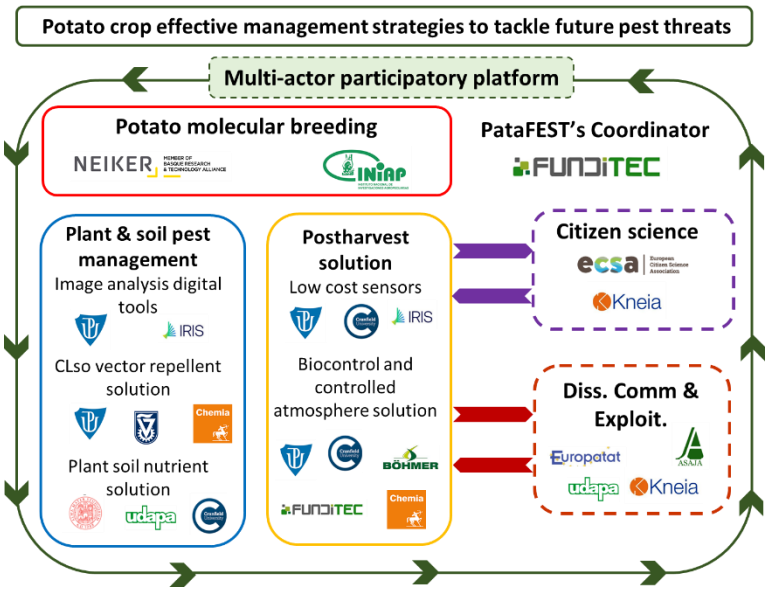

Figure 1. Overall concept of PataFEST

[6] https://gd.eppo.int/taxon/PARZCO/distribution. Last updated 2021-10-05
[7] Mora, V. et al. (2021). Frontiers in Microbiol. 12: 700663.
[8] PM 1/1 (30) EPPO A1 and A2. Lists of pests recommended for regulation as quarantine pests. September 2021.
[9] Powelson, M.L., Rowe, R.C. (2008). In: Johnson D., ed. Potato Health Management, 2nd edn. St Paul, MN, USA: APS Press, 183-195.
[10] Knowles N.R., Plissey E.S. (2008). In: Johnson D., ed. Potato Health Management, 2nd edn. St Paul, MN, USA: APS Press, 79–99.

develop sustainable and effective **plant and soil treatments** against the CLso vector and soil-borne pathogens combined with other innovative technological solutions of image analysis digital tools that allow the early detection of pest threat and **3)** to develop and validate **postharvest technologies** (biocontrol coating solution, controlled atmosphere storage technologies and high-performance volatile organic compounds (VOCs) sensors) to control the incidence of soil pathogens and maintain the quality of potato tubers stored. All these strategies and technologies will be incorporated into an **integrated pest management (IPM) approach** to evaluate their performance in a relevant environment (TRL5). PataFEST will ensure a long-term strategy for tackling future potato pest threats (Fig. 1).

PataFEST strategies are aligned with the topic "Horizon-CL6-2022-FARM2FORK-02-02 two stages: Emerging and future risks to plant health" promoting the transition to a fair, healthy and resilient agriculture and including alternative and sustainable technologies to reduce the use of chemicals both in field and in postharvest application. The project proposes **5 objectives (O)** with their respective **5 outcomes (OC)**, articulated under **15 results (R).**

O1. To develop a unique approach to identified resistance genes in potato varieties against pest and pathogens selected. Including: 1/ to setup a comprehensive catalogue of mapped resistance genes against CLso pest disease and soil borne postharvest pathogens; 2/ to determine molecular characterization of the working collection of potato accessions; 3/ to validate the resistance genes in the working collection of potato accessions.

R1	Full identification, compilation, and mapping against the reference genome of all annotated potato RGs, QTLs and orthologous genes from the Plant Resistance Genes database (PRGdb).
R2	A broad germplasm working collection will be analysed (**40 wild species**, **35 native varieties** and **230 old and new** European cultivars). In addition, **100 samples** from **Ecuador** (advanced breeding materials, wild species, native varieties) which have been characterized for their tolerance level to both, CLso and its insect vector.
R3	Identification of promising resistance gene SNP for targeted diseases and pests which can be applied in marker-assisted selection within molecular potato breeding.
OC1	PataFEST project will provide relevant and free access information to better identify the most suitable potato varieties resistant to plant pest threat and soil-borne pathogens in postharvest storage.
O1's pertinence to the call requirements: *"Increase knowledge of the biology, pathways of entry, behaviour in the plant-soil system..."; "Identify and introduce resistance traits to support the long-term sustainability of agriculture and forestry in the EU and/or Associated Countries".*	

O2. To develop innovative strategies to control potato pest spread and reduce potato postharvest disease. Including: 1/ to develop low-cost diagnostic solutions as VOCs sensor devices and image analysis (e.g. smartphone) application for the early detection and surveillance of the plant pest disease and postharvest pathogens in potato warehouses; 2/ to provide natural and sustainable solutions to prevent pest spread by controlling the vector in potato plants and investigating biocontrol and controlled atmosphere strategies to reduce postharvest disease incidence and severity; 3/ to reduce the incidence of CLso and post-harvest pathogens in potato crop increasing plant health status thanks to the optimised nutrient availability and improved plant soil management.

R4	Identification of phenotyping traits (biomarkers) in infected plants and tubers through Phenotype-based High-throughput Screening (PHTS) which uses different non-invasive sensors as RGB, FluorCam, IR, and hyperspectral cameras. The best identified biomarkers will be selected and use as a model for the **development** of "low cost", user-friendly and robust **digital tool** (smartphone app) and **portable sensor devices** for the **early identification** of the disease in both potato plants and tubers. Moreover, the PHTS will contribute to identifying solutions (compounds) that can help deal with the disease both in plant health and during postharvest.
R5	Non-toxic and sustainable **wax sprays** will render potato leaves super hydrophobicity providing **anti-adhesive** performance against CLso bacteria vector (psyllid) and **anti-pathogenic properties**.
R6	Enzyme and antagonistic yeasts-based biocontrol coatings will be tested on potato tubers stored under different graduated CO_2 and O_2 controlled atmosphere (CA) conditions to **decrease** at least **40-50% the incidence of** some important **postharvest pathogen diseases** selected as dry rot (*F. sambucinum)*, silver scurf and black dot *(C. coccodes,* and *H. solani)* and ensure consumer safety.
R7	Chitin and calcium carbonate-based **alternative biofertilizers** obtained from the valorisation of aquaculture biomass residues will be formulated as carriers of nutrient elements reducing susceptibility of potato plant to pathogens incidence both at foliar and tuber level. Soil management task will also include field trials to **evaluate pest pathway** of entry and **drivers of spread** as **future climate scenarios** (flooding, coldness, drought, etc.) and ecological degradation processes (modification of soil organic matter, etc.).
OC2	Use of early detection tools, sustainable solutions to improve plant and soil health and control postharvest pathogens will be helpful for meaningful reduction of pesticide and fungicide usage in potato crops.
O2's pertinence to the call requirements: *"Develop and uptake rapid and effective tools for the prevention of entry, spread and establishment, early detection, surveillance, treatment and (bio) control of plant pests; "influence of climate change and globalisation of pest(s)".*	

O3. To validate the Integrated Pest Management. Including:1/ to implement the technologies developed and carry out plant bioassays trials against CLso pest disease in an intended environments beyond Europe (e.g., Ecuador); 2/ to validate the postharvest technology and soil treatment solution at TRL5.

R8	Potato varieties identified and selected for their resistant traits against CLso and its vector will be tested in field bioassay trials under real condition (potato plants affected by the bacteria in Ecuadorian geographical areas). Digital tool developed will be operated from a mobile hardware unit (smartphone or tablet) and validated in field to diagnose the presence of CLso disease in potato crops.
R9	Pesticide free natural solution performance will be formulated at preindustrial level and evaluated on potato plants (**2 hectares**) to study the repellent effectiveness against BC in Ecuadorian geographical areas.
R10	Biocontrol naturals solution prototype will be formulated at pre-industrial level and tested on **2 tons** of tubers stored under optimized CO_2 and O_2 controlled atmosphere to control postharvest pathogen disease as F. *sambucinum*, *C. coccodes*, and *H. solani*. Sensors developed will be tested in potato warehouses under real storage conditions. Trails will be carried out in warehouses located in Spain and Germany.
R11	Soil treatment solution based on aquaculture by-product biomass residues will be evaluated in **1 hectare** of potato crop fields under real cultivation conditions in Spain.
OC3	Carry out trials both in potato plant and in tubers during postharvest storage of the technologies developed at TRL5 will validate the effectiveness of the integration pest management to control emerging threats and disease for potato crops.
O3's pertinence to the call requirements: *"Sustainable and integrated pest management"; "International cooperation"; "Economic implication for farmer and ecological impacts of plant pests"*.	

O4. To engage citizen as a source to monitor pests' disease. Including: 1/ to organize citizen science programmes to collect information database related to potato pest disease; 2/ to use digital tools and citizen science online platform.

R12	Involving individuals (citizen scientists) in such programmes will provide a low-cost means of maximizing search efforts across regions.
R13	**Three** types of **citizen science activities**: 1) Open source web activities in existing platforms such as SciStarter.org and/or eu-citizen.science; 2) smart-phone apps to collect ecology of plant pests (e.g. pictures of potential threats); 3) Citizen and farmers surveys, training and testing programmes.
OC4	The use of citizen science in the agriculture sector will be a fruitful opportunity to provide an effective supplement to pest management approaches.
O4's pertinence to the call requirements: *"Integrate citizen science as a tool to monitor emerging pests"*.	

O5. To create a multi-actor participatory platform. Including: 1/ to involve the scientific & technical community, companies, and citizens/consumers across the potato value chain; 2/ to create a public participatory platform ensuring the effective dissemination of project results and outcomes to appropriate target audiences.

R14	Creation of multi-actor participatory platform to share knowledge, expertise, and background across the interdisciplinary consortium to generate at least **40 "practice abstracts"** to contribute to EIP-AGRI repository.
R15	Project dissemination results will be shared through relevant channels to support the impact and exploitation outcomes including websites, social media, training materials, social events, etc.
OC5	Providing the foundations and drivers for future development and implementation of IPM strategies by farmers and other key players in the event that these emerging potato pests outbreak in EU.
O5's pertinence to the call requirements: *"Proposals must implement the "multi-actor approach" including a range of actors to ensure that knowledge and needs from various sectors such as research, plant health services and the farming/forestry sector are brought together"*.	

1.1.2 Ambition and progress beyond the state of art

1.1.2.1 Ambition and innovation potential

PataFEST project aims to provide a powerful European initiative for a sustainable IPM strategies to treat and control the presence of CLso pest and its vector in plant and the incidence of several soil-borne pathogens in the potato postharvest storage. Beyond the state of art, several innovative strategies will be implemented within the project:

Molecular breeding: Large part of the information is available but scattered over different sources will be combined and connected in one singular approach. For the first time, an innovative experimental association mapping regarding the ZC disease will be carried out using novel enriched base sequencing methods as Single Primer Enrichment (SPET) and sequence probe capture technologies. Furthermore, the generated combined information obtained in PataFEST will represent a universal and powerful tool for analysing any new emerging pest or disease in potato in the future, by simply applying additional validations of potential R-gene SNP.

Early detection tools: PHTS technology will be adapted to identify potato phenotype treats according to changes of selected biomarkers provoked by stress-inducing factors in plant models. By this way, it will be possible to diagnose

with more accuracy and in real time the potato plant health and select the most suitable methodology and sensor for the developing of a prototype tool able to perform an early detection of a pest or a pathogen.

Pests control solutions: The development of sustainable and innovative pre and postharvest solutions based on environmental-friendly ingredients and biotechnological agents will be a valid alternative of the excessive chemical compounds use in potato crops. PataFEST solutions will mainly provide protection against pest and pathogens, however their application will contribute to reduce respiration rate and dehydration both in potato plant and tubers.

Plant soil management: The innovation of the solution proposed in PataFEST will consist in the value improving of biomass residues by-products such chitin and calcium carbonate not used directly as fertiliser but as a carrier able to deliver and improve availability of essential elements to potato plants. Besides, in PataFEST testing different soil conditions will deliver understanding of how manage future climate scenarios on disease development to be tested at relevant scales, yet under replicable conditions.

Integrated pest management: PataFEST project will promote the opportunity to evaluate the synergy of the technological solutions within IPM strategies both for potato plant and postharvest tubers at TRL5. Field and postharvest trials will involve the participation of potato packers, cooperative, trader and farmer associations.

1.1.2.2 Positioning and Technological Readiness Level (TRL)

Innovation & Partners Involved	TRL initial → final	Justify current and final TRL
Potato germplasm collection (NEIK, INIAP)	2→5	The available published data will be converged in one complete collection data base and a singular validation will be performed using novel targeting sequencing methods (SPET and Capture).
Developments of tools for the pest and pathogen disease early detection (UPOL, IRIS, CRUN)	2→5	PHTS as a non-invasive real-time analysis has been used in several plant models. PataFEST research activities will allow to correlate potato phenotype and stress-inducing factors (e.g., study elicitors of biotic stress) like those provoked by pests and pathogens selected. The identification of specific potato health biomarkers (including volatiles) will allow developing high-performance sensors able to provide a fast and early- stage detection of emerging diseases.
Postharvest biocontrol solution (FUND)	3→5	Biocontrol solutions have been previously tested on surface of other fruits and vegetables against pathogens as *Colletothricum spp*. In PataFEST the solutions will be tested against soil-borne pathogens affecting potato crops.
Plant protective wax solution (TECHN)	3→5	Multi-functional pesticide-free treatment has been successfully tested on plant leaves controlling pathogens and bacteria disease. In PataFEST wax-based formulation will be tested as foliar spray solution to repel CLso vector, inhibit of soil-borne pathogen and control plant dehydration.
Plant soil sustainable solution (UNIBO)	3→5	Seashells biomass residues have been already tested for the formulation of new advanced materials for other industrial applications. In PataFEST innovative formulations will be tested to improve the soil quality and plant health in potato crops.

1.2 Methodology

1.2.1 Concept and overall methodology

PataFEST is a 48-month RIA action project based on the following activities against the presence of CLso pest threat in potato plants and pathogens incidences in postharvest tubers.

- **Molecular breeding:** NEIK will establish a singular and complete catalogue of mapped resistance genes against potato pest and diseases using different tools, starting with the compilation of all already annotated and published RGs on the potato reference genome. This initial compilation will be completed by mapping recently published potato resistance genes to the Potato Reference Genome. Besides, this collection will be extended by the compilation of all relevant resistance genes from de PRGdb, searching the potato orthologues; mapping and validating these genes to the Reference Potato Genome. Furthermore, published Potato Resistance QTLs will be mapped to the reference genome via projection, using collocated probes/primers with known sequences and identifying the potential collocated RG that caused the QTL. In a second step, a molecular characterization of the working collection of potato accessions will be achieved using SNP GGP V3 Potato array, which contains already many resistance genes. Additional important RG will be analysed for SNP by using innovative and novel targeting sequencing strategies such as SPET or CAPTURE with the Germplasm accessions. Finally, NEIK and INIAP will validate the RG by characterizing the accessions of the Germplasm Collection with respect to their resistance level to important diseases (and pests) using field trails and bioassays. Moreover, Association Studies between observed phenotypic resistance levels and molecular SNP data will be performed for validating important resistance genes.
- **Early detection and surveillance tools:** UPOL will perform correlation of induced plant stress conditions with data obtained by a photometric system used in PHTS approaches. The system based on RGB cameras, FluorCam,

IR, and hyperspectral cameras will allow the study of the phenotyping changes of plant and tubers in non-invasive manner. The studies will be performed in model plants under in vitro and in soil testing in crop-based bioassays confirming the possibility of a biological translation through species. The developed screening approaches will identify and describe the plant response allowing the identification of new targeted biomarkers. A sequence of serial and parallel targeted bioassays based on direct antibacterial and antifungal in vitro testing will be combined with high-throughput plant-based bioassays using different model pathosystems. Complementarily, CRUN will identify VOCs related to the infection of potato tubers with the selected pathogens. For this, the headspace of small potato microcosm experiments will be analysed using chromatography techniques as it allows a high concentration fold of the compounds produced. A full metabolomic approach will done by means of LC-MS/MS to identify non-volatile biomarkers, such as phenolic compounds, non-structural carbohydrates, and plant growth regulators, linked to the development of the pathogens during potato postharvest storage.

On the other side, IRIS will build up a multilayer digital tool (smartphone or tablet app) which will act as a storage unit and data processing unit for the image analysis. The inserted data will be processed within the platform to generate an accurate response in terms of detection. In addition, artificial intelligence-based models also developed by IRIS will be embedded in the platform to generate results based on the processing of the data captured by the operator. The digital tool will allow remote access as well as it will be programmed as a self-learning database, meaning that a "low-cost" device could be used for early detections of the desired markers.

- **Pest and pathogen disease control solutions:** TECHN and FUND will work on the development of chemical-free solutions to treat and control plant pests and postharvest pathogens disease, respectively. TECHN will propose to develop non-toxic and sustainable wax sprays amended with bioactive natural compounds (e.g., essential oils) to render potato plant leaves super hydrophobicity properties and provide both repellent and anti-adhesive performance against CLso vector and soil-borne antifungal effect. FUND will formulate water-based biocontrol coatings for postharvest potato tubers. Enzymes as lactoperoxidase system (LPS) and antagonist yeasts as *W. anomalus* will be selected as biocontrol agents and incorporated into formulation. Moreover, hydrocolloids and lipids will be also part of the formulation to provide high oxygen and water vapour barrier properties to maintain the quality of potato stored. Besides, UPOL will carry out simultaneous and cross-testing bioassays by the application of PHTS technology to provide a selection of tolerance increasing formulations against both plant pest and postharvest pathogens selected. CHEM as an expert industrial developer of crop protection products, will be in charge to manufacture both solutions at pre-industrial scale (TRL5) and evaluate the feasibility of their industrial exploitation.

  CRUN will work on the respiratory gases (CO_2 and O_2) during long-term postharvest storage; tubers will be subjected to different graduated CA environments and relative humidity to investigate the potential of this technology to reduce the disease development under commercial conditions.
- **Soil pest management strategies:** UNIBO will obtain carriers from biomasses residues that will be functionalised with several elements and applied directly to the soil in experimental plots under controlled conditions (pots in greenhouse). The trials will evaluate the effect of treatments on soil elements bioavailability and microelements contents in leaves and tuber. Moreover, the innovative fertilisation will be tested under CLso infection evaluating the titre of pathogen in potato tissues by molecular analysis using Differential-Display Reverse Transcription-PCR (DDRT-PCR) approach. Tuber produced under the described experimental conditions will be also assessed for resistance to postharvest pathogens. Besides, disease incident trials under different climate scenarios mimicking predicted extreme rainfall and drought periods will be conducted at CRUN in a phenotyping glasshouse facility designed to trial crop-soil systems at pilot scale under a multi-sensor gantry. The facility replicates all stages of the whole crop production cycle plus rotational context (multiple cropping seasons) with predefined soil physical conditions affecting fungal and bacterial movement, and recreates the dynamic interrelationships of soil health, water use, biotic stresses (pathogens, weeds, pests), environmental variables.
- **Integrated plant management:** Several IPM strategies will be considered in PataFEST project. One case of study will be carried out in Ecuador. There, INIAP will host field bioassay trials in their facilities under real condition (presence of CLso and its vector in potato plant) evaluating the behaviour of potato varieties already selected for their resistant traits against CLso and its vector. Besides, INIAP will validated in potato plant the capacity of the image analysis digital tools to predict the presence of CLso disease in potato plants in fields. Finally, pesticide-free natural solution performance will be evaluated on potato plant to study the repellent effectiveness against BC presence and the antifungal properties against soil-borne pathogen in tubers harvested. In Europe, two IPM cases of study will be organized both in UDAPA (Spain) and in BOEH (Germany) potato warehouses for the postharvest technological solutions trials. Biocontrol superficial coating solution combined with CA technology and low-cost sensor device will be tested to early detect and control the pathogen spoilage in potato stored. Finally, UDAPA will also evaluated aquaculture by-product-based soil treatment application in potato crop under real cultivation conditions (Spain). Other case of studies for soil management and postharvest IPM strategies will be planned around Europe thanks to the potential involvement of ASAJA and EUPAT associated members.

- **Citizen science:** ECSA and KNEIA will be in charge to involve citizen scientist in the activities that will be organised along the PataFEST project execution. Results obtained will significantly contribute to boost the collaborative efforts to detect and control emerging pest threat and pathogens disease incidence.
- **Joint Research Center involvement:** As this topic describes, the involvement of the JRC is envisaged once the project is funded. JRC will support the analysis of the potential impact of the studied pests and the development of economic and environmental solutions.

1.2.2 Related research and innovation activities

The activities developed in PataFEST linked with many previous and ongoing R&D initiatives described below:

Project	Main result/knowledge feeding into PataFEST
PAPACLIMA (FAO, 2016-2019). Partner: **NEIK**	Detection of candidate genes against fungi. Knowledge and software for performing different association mapping techniques Evaluation of fungal diseases
POISE (EIG concert Japan, 2016). Partner: **UPOL**	Identification of biomarkers related to tolerance level to abiotic stress in tomato and rice plant.
QCAP: (Interreg NWE, 2017-2019). Partner: **CRUN**	Detection and identification of potato spoilage VOC biomarkers under postharvest cold storage conditions.
PHOTONFOOD (H2020, 2021-2024). Partner: **IRIS**	Development of the global system control, software and data management and processing platform.
ANTISEPTIC-WAX (ERC H2020-PoC, 2018-2020). Partner: **TECHN**	Environmentally friendly wax-spray green technology to eliminate attachment and development of pests on plant leaves surfaces.
REPEAR (FP7, 2014-2017). Partner: **FUND**	Development of superficial coatings based on natural additive to reduce respiration rate, control dehydration, and pathogens spoilage of postharvest pears.
ADAPT (H2020, 2020-2024). Partner: **EUPAT**	Dissemination of the project objectives and results via internal and external communication channels.

1.2.3 Gender dimension

PataFEST does not include the integration of the **gender dimension (sex and gender analysis) in R&I as the topic 2022-F2F-02-02 does not require it**. Nevertheless, the project has a strong multi-actor participatory approach and involvement of citizen, thus it will commit to address the diverse needs of citizens regardless their gender and sex.

1.2.4 Open science practices

The project will follow an Open Science Strategy (OSS). The preliminary type of research data, exploitation route and IP strategy will be outlined in the Exploitation Plan (available at the 2nd stage). This plan will show that some results with commercial and/or industrial important value will be protected as IPR or trade secrets, and thus they will not be included in PataFEST's OSS. The research results obtained by the work carried out by researchers will follow these Open Science practices: **1) Open Access (OA) to peer-reviewed scientific publications** ensuring at least under upon publication, deposition of the author accepted manuscript (AAMs) or version of record (VOR) in a **trusted repository** and immediate open access via the repository under CC BY or equivalent. Alternatively, some research results will be made available in the **Open Research Europe**; 2) **Early and open sharing of research** through pre-registration, registered reports, pre-prints or crowd-sourcing; 3) Researchers will participate as much as possible in **open peer-review** in related topic of PataFEST; **4) Multi-actor approach activities** in the co-creation of tasks from the beginning and across the project duration.

1.2.5 Research data management and other research outputs

The Data Management Plan (DMP) will consider all the data that the project will generate and use and will detail whether and how it will be used, exploited or made accessible for verification and re-use; how and under what conditions it will be curated; and how it will be interoperable with regard to specific quality standards. Data refers to any kind of result including statistics, results of experiments, measurements, observations resulting from trials and demonstration, survey results, interview recordings, images and bibliographic metadata used for the validation publications. According to EC recommendation, as part of making data findable, accessible, interoperable and re-usable (FAIR), the DMP will include information on: **1) handling of data during & after** the end of PataFEST project; **2) what data** will be collected, processed and/or generated; **3)** which **methodology and standards** will be applied; **4)** whether data will be **shared/made in OA**; **5)** how data will be **curated and preserved** (including post-project). DMP will not have impact on IPR, because the decision to publish OA documents come after a more general decision on whether to first seek protection for IPR. The data will be deposited via OA in a trusted repository under the latest available version of the Creative Commons Attribution International Public License (CCBY) or Creative Commons Public Domain Dedication (CC0) or a licence with equivalent rights.

1.2.6 Compliance with the ´do no significant harm principle (DNSH)'

All the research activities of PataFEST will comply with DNSH principle under the Recovery and Resilience Facility Regulation of EU. PataFEST fulfils DNSH in different ways: 1) no activity is harming the **climate change mitigation** as the developed plant health products are fossil-free; 2) no activity is harming the **circular economy** concept. All the contrary, the project will apply circularity principles in the plant health protection products leading to more resource efficiency, to zero-waste practices, and to sustainable and 4) no activity is harming **biodiversity and ecosystems** as the project enhances potato genetic variety on agricultural land.

2 IMPACT

2.1 Project's pathways towards impact

This section covers PataFEST's impact pathways towards the **Expected Outcomes (EOs)** at medium-term (from project's end +4 years later)[11] and on the **Expected Impacts (EI)**[12] at long-term. These pathways are defined by clear, measurable, quantified, and assessable **impact indicators** that fall under the following **impact category**: scientific (**Scie-I**), societal (**Soc-I**) and economic & technological impacts (**Econ&Tech-I**). The results of the project will be used or further up-take by direct **target groups** (Fig 3). PataFEST will generate credible and sustainable impact pathways mainly in 5 impact areas: *'Sustainable food systems from farm to fork on land*, '*Enhancing ecosystems and biodiversity on land', 'Clean Soil',* '*A resilient EU prepared for emerging threats*' and '*Inclusive growth and new job opportunities*' in line with some of the 9 **Key Impact Pathways** (**KIPs**) defined within the **Key Strategic Orientation KSO B** '*Restoring Europe's ecosystems and biodiversity, and managing sustainably natural resources to ensure food security and a clean and healthy environment'* of the HEU Strategic Plan.

Pathway towards Impact 1: Increase the molecular and biological information of potato with a thorough mapping of its resistance genes in response of emerging pests under current climate change. *Corresponding to project´s objective O1.*

Impact Indicator 1: Acquisition of data related to potato resistance genes → *Impact category*: **Scie-I**

As explained in "Excellence", ZC disease is both costly and difficult to manage. Application of chemical pesticides has been the primary method of combating ZC in numerous regions around the world. Unfortunately, overuse of pesticides has resulted in ZC-vector resistance. As a result, pesticide use is both economically and environmentally unsustainable, and goes against the **2030-EU Action Plan** target of lowering chemical pesticide use by 50% to achieve the "zero pollution" targets. PataFEST proposes a method for studying potato resistance genes to the ZC vector psyllid BC and/or CLso based on potato genetic resources. This will ensure a long-term sustainable and successful strategy for managing increasing ZC disease and cultivating more resistant potatoes. PataFEST will thus offer a strategy in line with the **EU Biodiversity Strategy 2030's** primary action to enhance genetic variety on agricultural land, which would benefit EU health and economy.

Scale and significance to Impact 1	Target Groups	HEU KIPs
Results-Short term: Identification and characterization of potato resistance genes against diseases caused by ZC and soil-borne pathogens (305 European samples and 100 samples from Ecuador amongst wild and native species and cultivars) *(linked to R1-R3).* *EO-medium term:* Complete map of potato genes resistance to emerging threats in EU. *EI-long term*: Prevent similar losses in potato production in the EU as those seen in other countries where these pests already exist.	Scientific/Technical Community Farmers Industry along the potato value chain Plant health policy makers	KIP 1 'Creating high-quality new knowledge'

Pathway towards Impact 2: Long-term sustainable potato pest and postharvest diseases integrated management, including prevention, early detection and surveillance strategies. *Corresponding to project´s objectives O2 and O3.*

Impact Indicator 2: Catalogue of innovative solutions to respond to new and emerging potato pests → *Impact category*: **Econ&Tech-I**

[11]The **Expected Outcomes (EO)** described in the topic **HORIZON-CL6-2022-F2F-02-02**: Emerging and future risks to plant health are:
EO 1*: Find adequate responses to new and/or emerging plant pests.*
EO 2*: Improve the understanding of drivers of plant pest emergence including the influence of climate change, ecosystem degradation and globalisation.*
EO 3*: Develop economic and environmentally sound solutions for effective pest management in farming and forestry in line with the principles of integrated pest management.*
EO 4*: Support to relevant plant health policies in the EU and Associated Countries.*

[12] PataFEST sets out the above pathways towards the following **Expected Impacts (IA)** specified in the destination 'Fair, healthy and environment-friendly food systems from primary production to consumption' of the Cluster 6 Work Programme:
EI 1: *sustainable, productive, climate-neutral and resilient farming systems […], while minimising pressure on ecosystems, restoring and enhancing biodiversity, […] generating fair economic returns for farmers.*
EI 3*: sustainable, healthy and inclusive food systems delivering co-benefits for climate mitigation and adaptation, […], sustainable healthy nutrition, safe food consumption, food poverty reduction, the inclusion of marginalised people, the empowerment of communities, and flourishing businesses.*

PataFEST's concept is structured around the **EU's agricultural research and innovation strategy**, notably the set of initiatives aimed at "**Priority 2: healthier plants**"[13]. As the EU strives to reduce the use of chemical pesticides[14], **biopesticides are being more widely used in agricultural practices** (biopesticides CAGR 15.4% versus synthetic pesticides CAGR 1.2% for the period 2018-2023), potentially introducing new pathogens not occurring before[15]. PataFEST will develop sustainable and systemic IPM strategies to address the prevention, monitoring, control, and management of ZC during cultivation and soil-borne pathogens during postharvest storage, as an alternative to the use of pesticides or other contentious methods, in combination with other innovative agronomic and digital solutions. In this regard, the project is fully in accordance with the actions of the **"Agriculture and the Green Deal"**, as it promotes the adoption of a long-term IPM by potato professionals to achieve a healthy and sustainable food system for people and the earth[16].

Scale and significance to Impact 2	Target Groups	HEU KIPs
Results-short term: integrate all the innovative solutions and validate the IPM strategies at TRL 5 in different geographical areas (Germany, Ecuador, Spain) (*linked to R4-R11*): a) bioassay field trials of selected potato varieties in Ecuador (*R8*). *b*) portable low-cost sensors and digital platform (*R4*); c*)* bio-based plant/tuber/soil health solutions (*R5-R7; R9-R11*). *EO-medium term:* laying the foundations for future development (TRL 6→9) of novel potato disease-strategies (e.g., bio-based chemical products stimulating the expressing of resistance genes). *EI-long term*: In the event that potato diseases enter in Europe, a rigorous, sustainable and economic Integrated Pest Management strategy might be implemented.	Farmers Bio-based industry of plant health solutions Industry along the potato value chain Citizens	KIP 7 'Generating innovation-based growth' KIP 9 'Leveraging investment in research and innovation'

Pathway towards Impact 3: Citizens and farmers as key actors for prevention and early detection of potential potato threats ('Citizen science'). *Corresponding to project´s objectives O4.*

Impact Indicator 3: Involvement of citizens across regions → *Impact category*: **Soc-I**

The potato IPM proposed in PataFEST will be **designed by and for individuals and farmers in the EU and Associated Countries**. Citizens will contribute to the studies in a variety of ways (e.g. expert reviews, potato photo submissions, training and testing programs, smartphone-collected information in apps, etc.) through citizen science activities inspired by other EU initiatives such as *eu-citizen.science* or the international citizen hub *'Science we can do together'*, with the goal of preventing and detecting potential threats to potato crops around the world and long-term adoption of the IPM. Everyone, regardless of age or professional background, will contribute to the project's research outcomes in order to have a significant impact not only during the project's implementation (2023-2026), but also in the medium and long term. The project will ensure low-cost and effective means of maximizing our outcomes across regions of the EU and internationally by involving citizens in the project from the beginning, throughout its duration, and beyond. Citizens will play a significant role in supplementing knowledge to technical pest management strategies.

Scale and significance to Impact 3	Target Groups	HEU KIPs
Results-short term: Research data collected and supported by the citizens in the project's research activities to monitor emerging potato pests → 3 types of citizen's activities *(linked to R12-R13).* *EO-medium term:* Share best-practices of the citizen science approach to other related pests. *EI-long term*: Adoption of citizen science systems for the surveillance and control of disease-carrying vectors by governments and private sector.	Citizens Farmers	KIP 6 'Strengthening the uptake of research and innovation in society'

Pathway towards Impact 4: Creation of an interactive innovation model to help the implementation of potato IPM across different professional backgrounds at local, regional, national and international scale ('Multi-Actor Approach'). *Corresponding to project´s objectives O5.*

Impact Indicator 4: "practice abstracts' of EIP-AGRI → *Impact category*: **Soc-I**

The **EIP-AGRI policy goal** of "speeding up" innovation will be met from the start of the project and in the long run thanks to the active participation of farmers and agri-food businesses through the key associations and cooperatives engaged (ASAJA, UDAPA, EUPAT). The **multi-actor participatory platform** will include the **scientific & technical community**, **companies** (SMEs, large industries, agricultural/potato associations, and cooperatives), and

[13] A strategic approach to EU agricultural research & innavation. Available here.
[14] Directive 2009/128/EC. Sustainable use of pesticides. Available here.
[15] Frost & Sullivan (2019). Global Crop Protection Market, Forecast to 2023.
[16] Integrated Pest Management (IPM) in the EU Green Deal. Available here.

citizens/consumers across the potato value chain (Fig. 2) to share knowledge and cooperate one with each other to maximize our results and outcomes.

This platform will be critical for enhancing rural community engagement at all scales (local, national, and international) and for disseminating the project's proposed solutions against emerging potato pests. PataFEST's impact will be amplified by close engagement with current **EIP Operational Groups** and other similar projects.

MULTI-ACTOR APPROACH across PataFEST consortium				
	Preharvest *Seed suppliers, improvement of varieties, plant health solutions*	**Production**	**Postharvest & Processing** *Handling, storage packaging*	**Distribution & Commercialisation**
Scientific/Technical Community	**NEIKER, INIAP, TECHN, UPOL, UNIBO**	**NEIKER**	**FUND, CRUN, UPOL**	
Farmers	**ASAJA, UDAPA**			
SMEs & Large industries	**CHEM, EUPAT, IRIS**	**EUPAT, UDAPA**	**IRIS, BOEH, CHEM, EUPAT, UDAPA**	**UDAPA, ASAJA**
Citizen Science & Consumers (ECSA, KNEIA) **Cooperatives & Associations (ASAJA, UDAPA, EUPAT)**				

Figure 2. Multi-actor approach applied in PataFEST.

Scale and significance to Impact 4	Target Groups	HEU KIPs
Results-short term: Share knowledge, experience, and background due to the complementary disciplines of the multiple actors → creation of **~40 "practice abstracts"** *(linked to R14-R15).* *EO-medium term:* acceptance and adoption of the PataFEST solutions by farmers and other actors of the potato value chain production. *EI-long term*: Implementation of potato IPM across all the actors → ensure economic sustainability of potato production, increase farm competitiveness and improve key actor's position in the value chain.	All actors of the potato value chain shown in Fig. 2	KIP 4 'Addressing EU policy priorities and global challenges through R&D' KIP 5 'Delivering benefits and impact through research and innovation missions'

All in all, PataFEST is build up with a strong commitment towards the key policies objectives of the **new Common Agricultural Policy (CAP)**, **Farm-to-Fork strategy** and **SDG (2, 8, 13, 15, 17)** for the period 2023-2027, briefly: 1) "Fostering knowledge & Innovations" → involving farmers to our research activities and outcomes; 2) "Halting & reversing biodiversity loss" → preserving more resistance potato varieties, 3) "Efficient natural resource management" → developing a sustainable, effective and long-term management strategies to tackle future potato pest; 4) "Contributing to climate change mitigation" → boosting sustainable and bio-based circular solutions for plant health (biopesticides from biomass sources); 5) "Increasing competitiveness" → involving all relevant actors in the potato value chain. The project also will support the "**Plant Health Law**" and all the associated implementing acts related to pests, commodities, plant health requirements, etc[17].

Requirements and potential barriers

Barrier/Requirement Description	Strategies for overcoming
Technology demand: low demand of the potato industry and/or bio-based industry of plant health solutions could undermine the further development of the solution at higher TRLs.	Scientific/Technical community of the project, in collaboration with other funded EU projects, initiatives and OPs of EIP-AGRI will create dedicated scientific forums to mitigate this potential barrier.
Lack of citizen engagement in the citizen science tool and poor adaption by farmers of IPM.	PataFEST will provide well documented results, dissemination materials and engagements activities to all the actors that will form a sound base to support our results and outcomes.
EPPO Standards associated to potato pests & guidelines[18]: PM1/002(30) EPPO A1 and A2 Lists of pests recommended for regulation as quarantine pests (2021); PM7/143(1) CLso; PM8/001(2) Potato; PM9/025(2) BC and CLso and the **guidelines in good plant protection practice** – **Potato** PP 2/2.	The novel potato management strategies for both pre- and postharvest will be carefully followed these standards in order to avoid any potential confrontation with existing standards.
EU Regulatory and Policy requirements: EU regulatory framework may affect production and commercialisation of potato pest management solutions. The Plant Health Regulation and EU Directive 2002/56/EC are key for the correct use of IPM to protect potato crops beyond project's duration.	The consortium, specifically our partners representing different actors across the potato value chain, will review the legislation applicable to protective measures against pests and other legislations applicable to PataFEST. Moreover, engagement with policy makers will be ensured.

[17] Plant Health Law (Regulation (EU) 2016/2031 and implementation acts are found here.
[18] All the documents related to EPPO standards for potato are available here.